

Northwind Resolve — One-Page Value Case

Business Problem

Northwind’s customer-service operations are fragmented across multiple systems, forcing agents to switch between tools, transfer complaints, manually investigate billing issues, and reconstruct customer context. This increases handling time, operating cost, and the risk of inconsistent resolutions.

Proposed Solution

Northwind Resolve provides a unified internal operations platform that combines complaint context, billing and meter information, case history, field-work requests, AI assistance, and tamper-evident audit records in one workflow.

The platform uses:

- **Gemini API** for structured case summaries, classification, recommendations, and customer-response drafts.
- **Deterministic backend rules** for billing-risk and business decisions.
- **PostgreSQL** for canonical operational context and analytics.
- **Solana Devnet** for tamper-evident hashes of critical events.

AI remains advisory; consequential actions remain controlled by backend rules and human operators.

Observed Baseline

Based on the supplied Northwind challenge data for the latest 12-month source period:

Metric	Observed Baseline
Complaints opened	13,709 / year
Complaints transferred between systems	4,392 / year
Manual billing corrections	3,945 / year
Average complaint handling cost	\$68 / complaint
Transferred complaint handling cost	\$121 / complaint

These are observed challenge-data inputs—not claimed savings.

Illustrative Scenario

Assume Northwind Resolve enables:

- **10% reduction in complaint volume**
- **20% reduction in transfers among remaining complaints**
- **\$100,000 annual operating cost**
- **\$250,000 implementation cost**

Projected complaint benefit: $13,709 \times 10\% \times \$68 = \$93,221.20$ / year

Incremental cost of a transferred complaint: $\$121 - \$68 = \$53$

Projected transfer benefit: $4,392 \times 90\% \times 20\% \times \$53 = \$41,899.68$ / year

Projected Value

Gross modeled benefit: \$135,120.88 / year

Projected annual net benefit: $\$135,120.88 - \$100,000 = \$35,120.88$ / year

Illustrative payback: $\$250,000 \div \$35,120.88 \times 12 \approx 85.4$ months

Why This Matters

The value case is intentionally transparent: operational baselines come from Northwind’s supplied data, finance unit costs are identified separately, and improvement percentages and implementation costs are explicit scenario assumptions. The platform does **not** present projected savings as realized savings.

The strongest value is therefore not only cost reduction, but a measurable operating model: fewer handoffs, faster case understanding, more consistent billing review, traceable AI assistance, and auditable critical actions.